

Capstone Report SecureWave Technology

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INFT1103: Operating System Security

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December 5, 2025



Introduction

Overview of SecureWave Technology

Microsoft Domain Environment

SecureWave uses Windows Server 2019 and Active Directory Domain Services for centralized network management and file sharing.

Security Vulnerabilities

Irregular updates, missing patch management policies, and outdated firmware increase risks of system instability and cyberattacks.



Overview of SecureWave Technology (Cont.)

Risk and Compliance Concerns

Security gaps can cause regulatory non-compliance and threaten operational continuity and data protection.

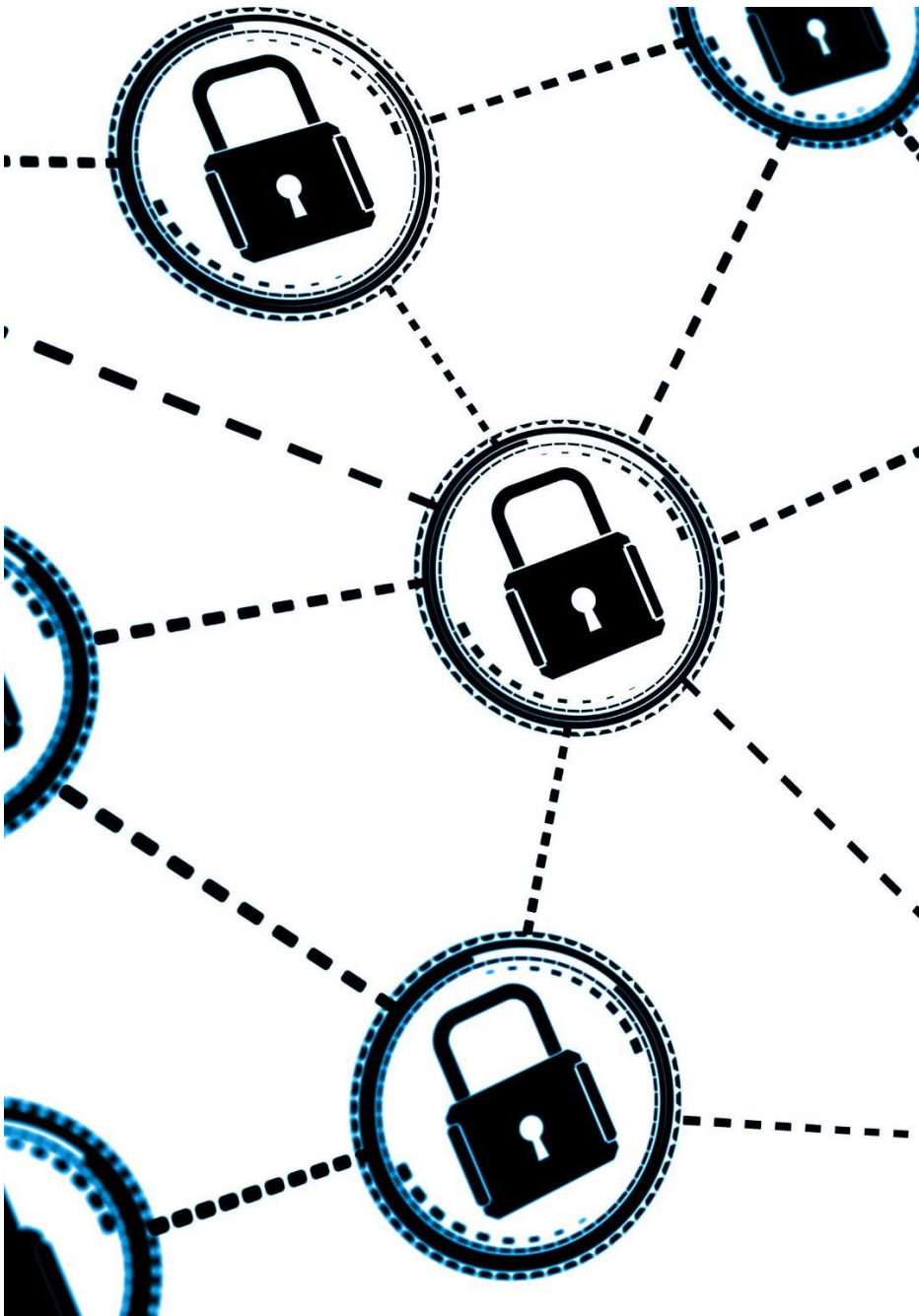
Need for Security Improvement

Addressing vulnerabilities is essential to strengthen SecureWave's security posture and protect sensitive information.



Security Vulnerability 1

Sporadic Patch Management



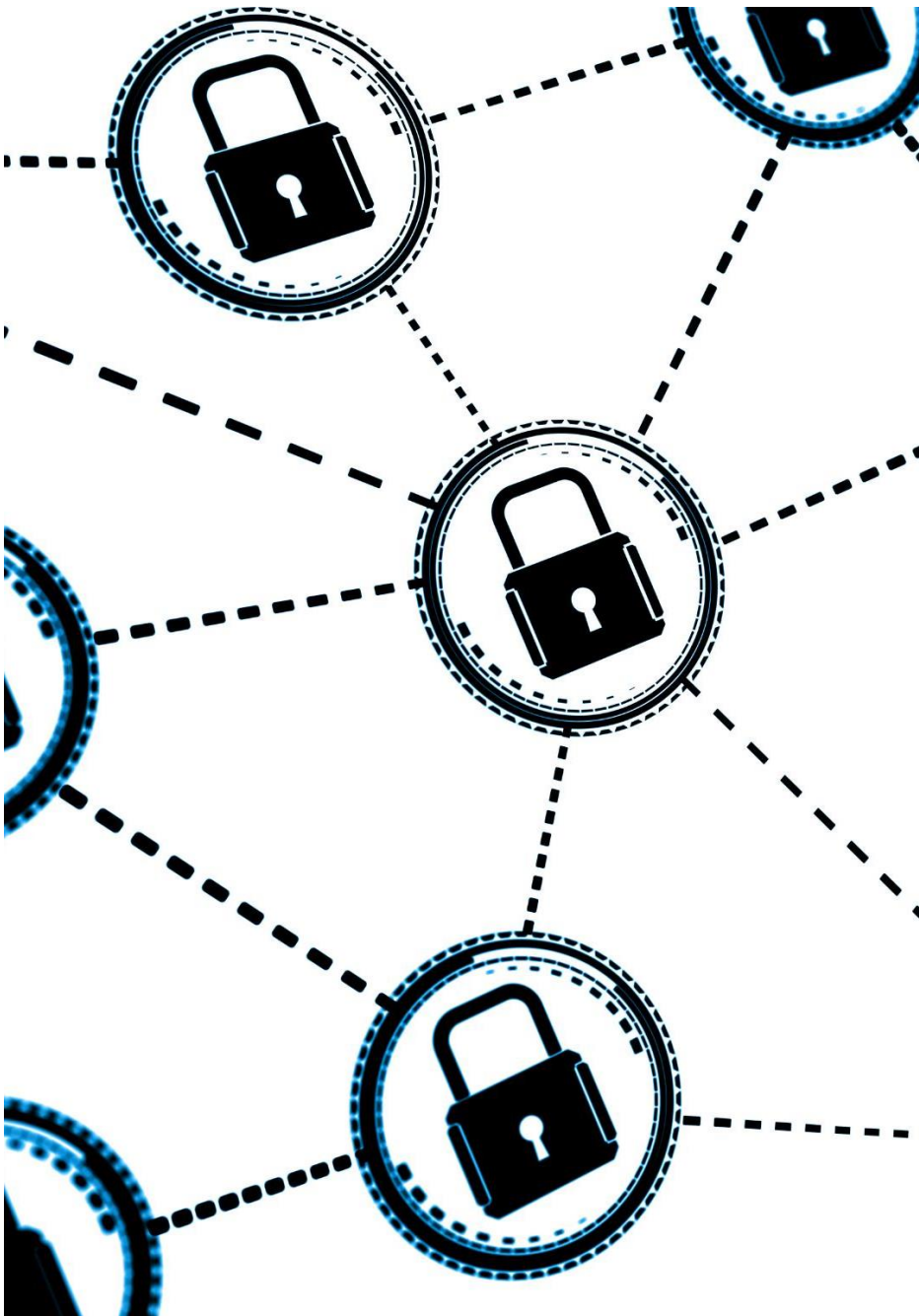
Sporadic Patch Management Risks

Inconsistent Patch Implementation

Lack of defined update policies leads to sporadic patching, increasing exposure to cybersecurity threats.

Operational and Compliance Risks

Unpatched systems cause instability, data corruption, and risk compliance violations with regulations like GDPR.



Sporadic Patch Management Risks (Cont.)

Technical Vulnerabilities

Exposed vulnerabilities such as CVE-2024-49113 enable remote code execution and domain controller compromises.

Network Security Weaknesses

Direct update server connections without WSUS can lead to man-in-the-middle attacks with misconfigured TLS proxies.

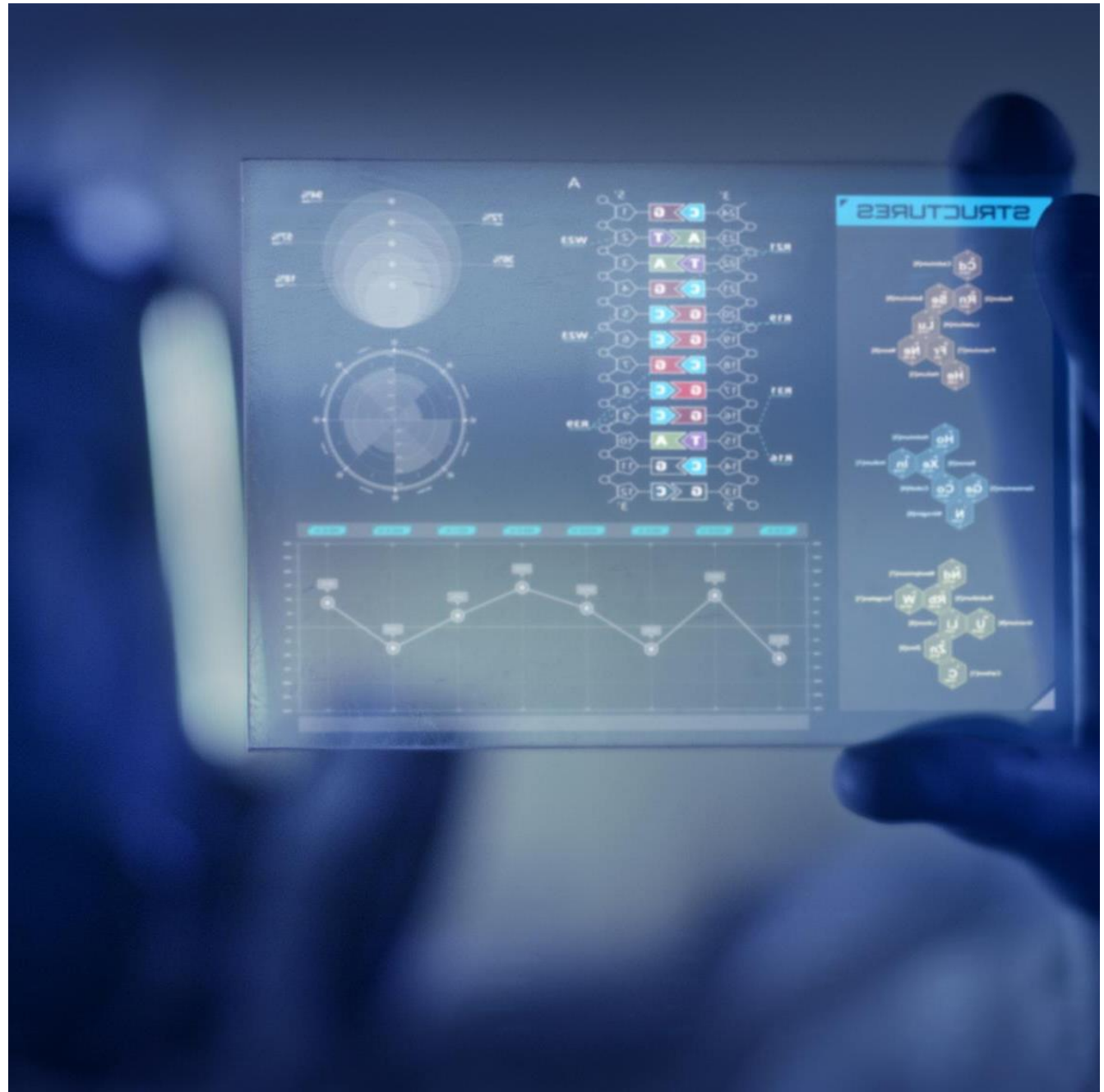
Addressing Patch Management Vulnerabilities

Routine System Updates

Implement routine updates across all systems to close security gaps and enhance system integrity effectively.

Vulnerability Scanning

Use tools like Nmap for regular scanning to detect outdated services and misconfigurations promptly.



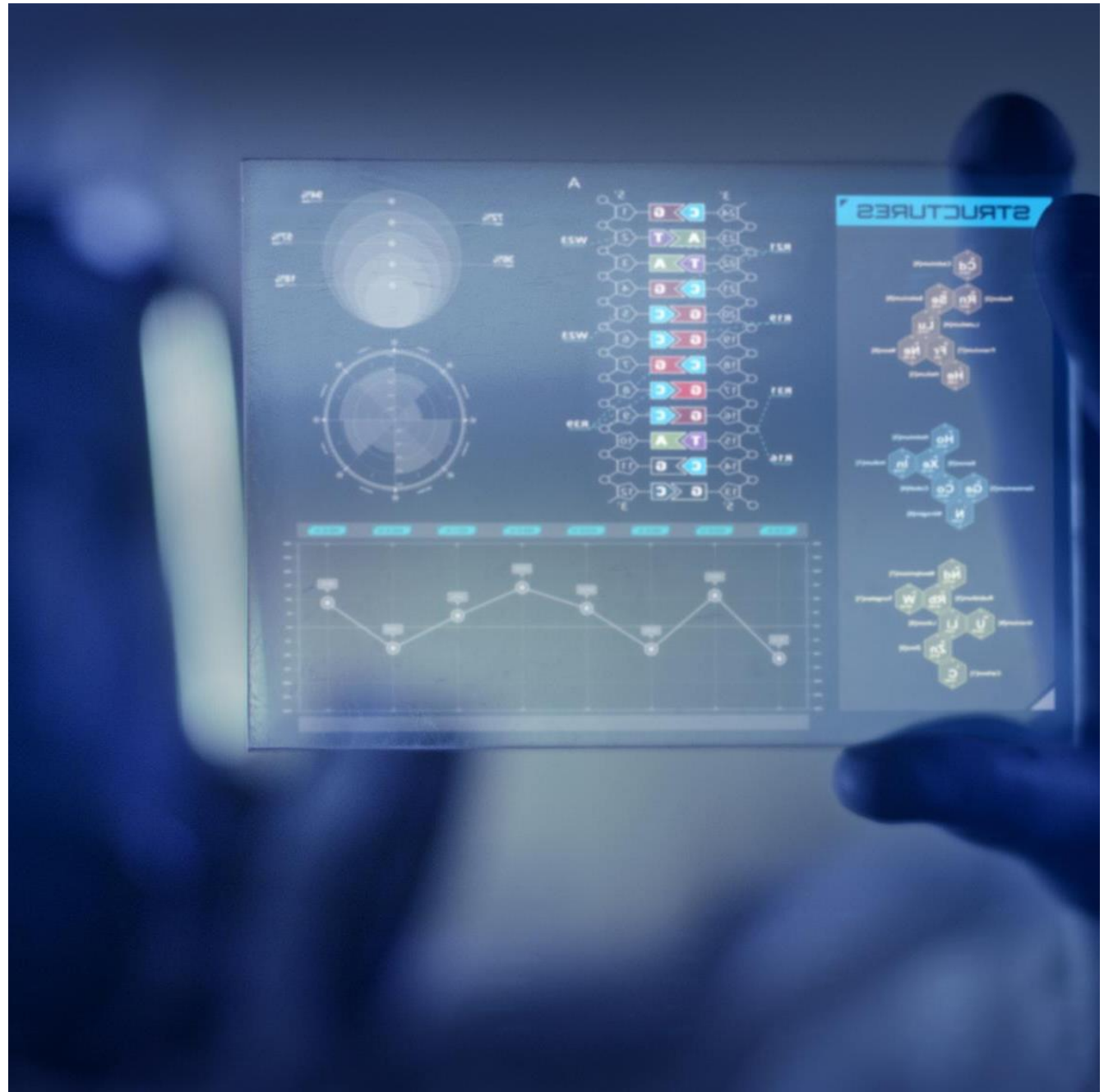
Addressing Patch Management Vulnerabilities (Cont.)

Patch Prioritization and Controls

Prioritize critical patches, especially for Windows Servers and domain controllers, and apply compensating controls for unpatched systems.

Integrated Security Solutions

Use advanced platforms integrating threat intelligence, vulnerability assessment, and patch management for efficient remediation.



Security Vulnerability 2

*Neglecting Firmware Updates on
Network Devices*

Neglecting Firmware Updates on Network Devices



Security Risks of Outdated Firmware

Outdated firmware can expose routers and switches to critical vulnerabilities allowing traffic rerouting and network compromise.

Impact on Network Performance

Obsolete firmware causes degraded network performance and compatibility issues with modern devices, risking outages.

Challenges for Security Monitoring

Security systems like SIEM may struggle to monitor traffic effectively due to unsupported protocols in outdated firmware.

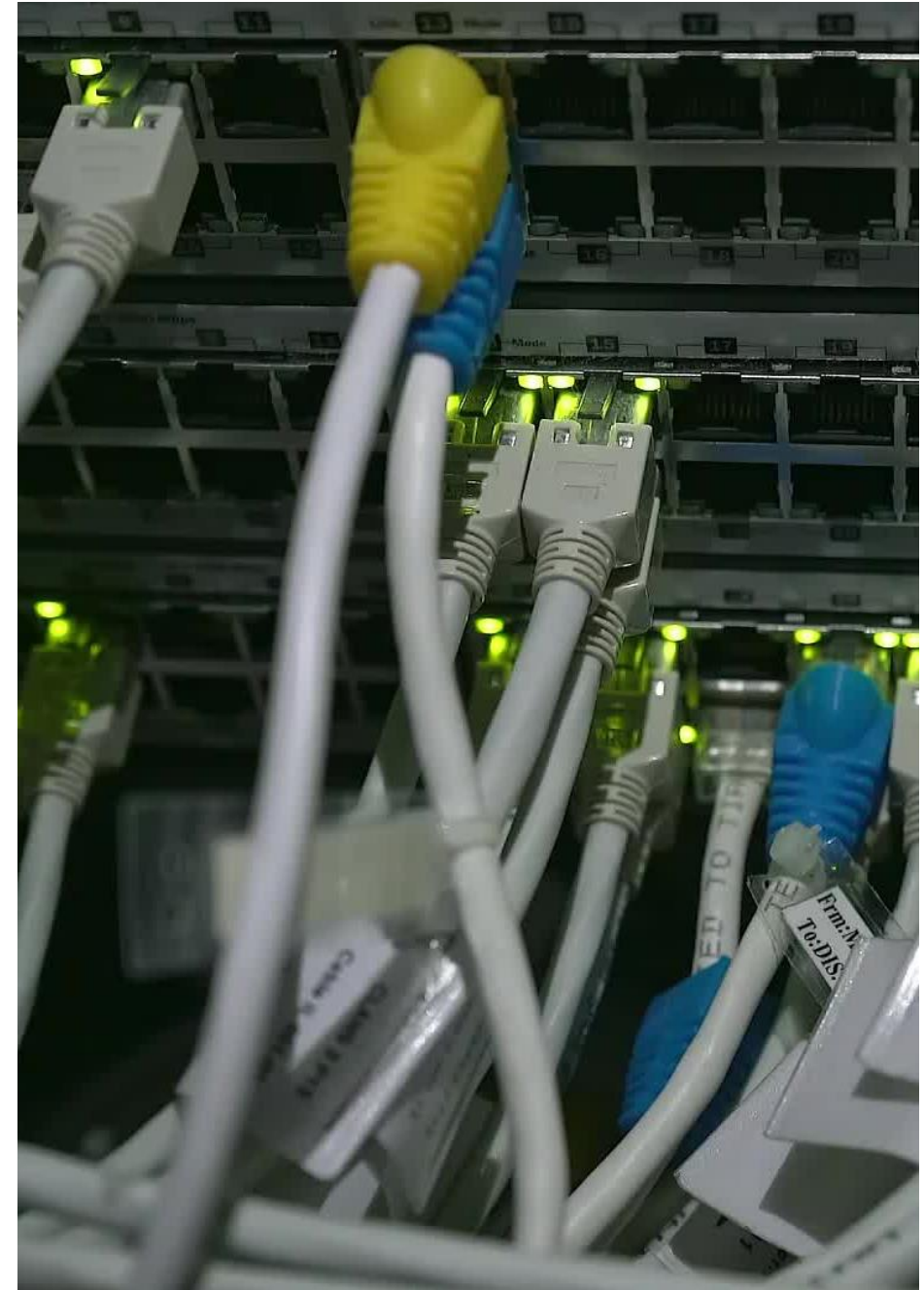
Addressing Firmware Vulnerabilities

Routine Firmware Updates

Regular firmware updates on routers and switches apply critical security patches and prevent vulnerabilities.

Device Inventory and Assessment

Maintaining an inventory and performing vulnerability assessments helps identify insecure devices promptly.



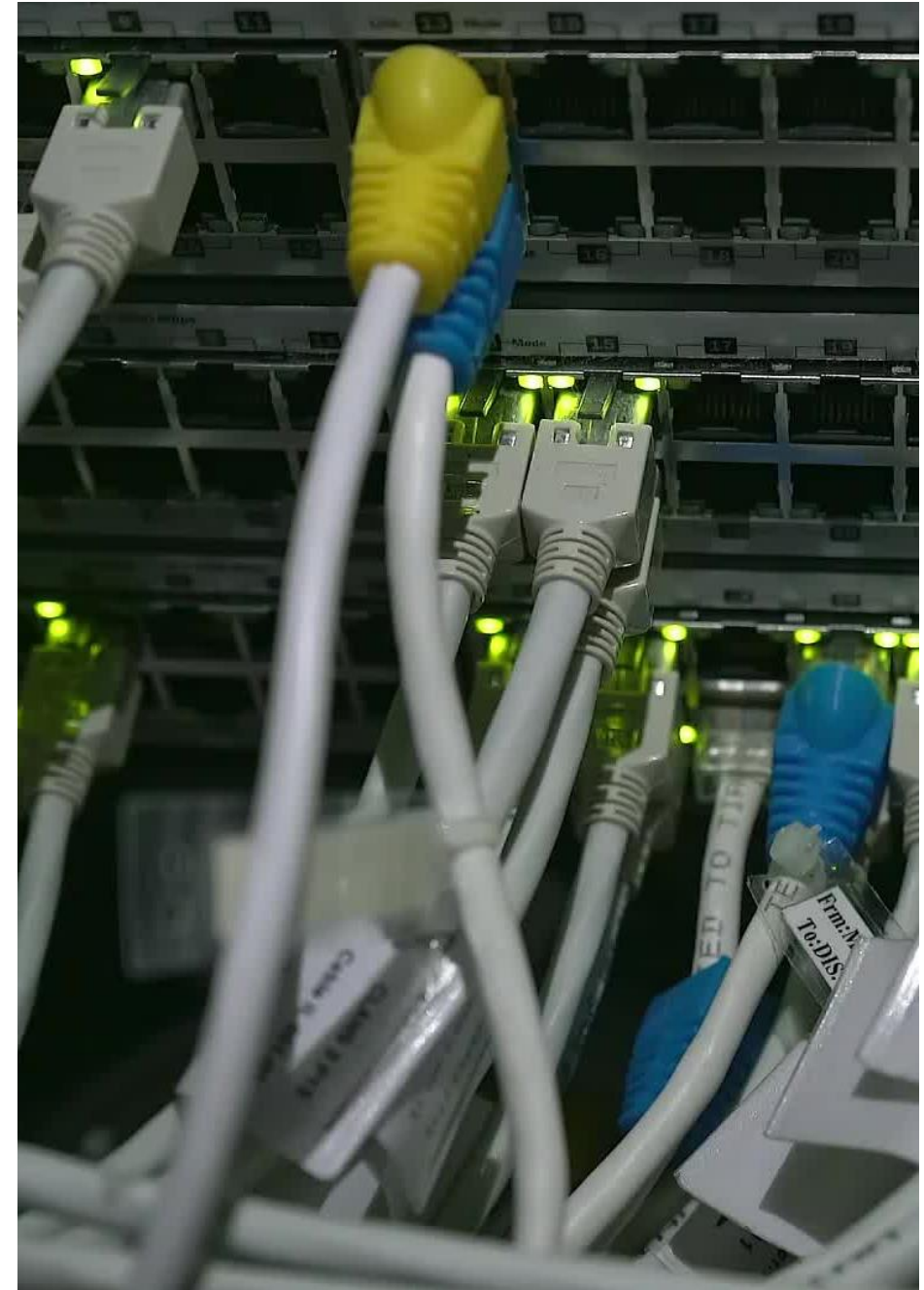
Addressing Firmware Vulnerabilities (Cont.)

Network Housekeeping

Consistent network housekeeping ensures compromised devices are replaced to maintain network security and performance.

Enhanced Security and Performance

Firmware updates improve security, compatibility, and performance, reducing bottlenecks and stabilizing connections.



Security Vulnerability 3

*Lack of Change Management and
Documentation*

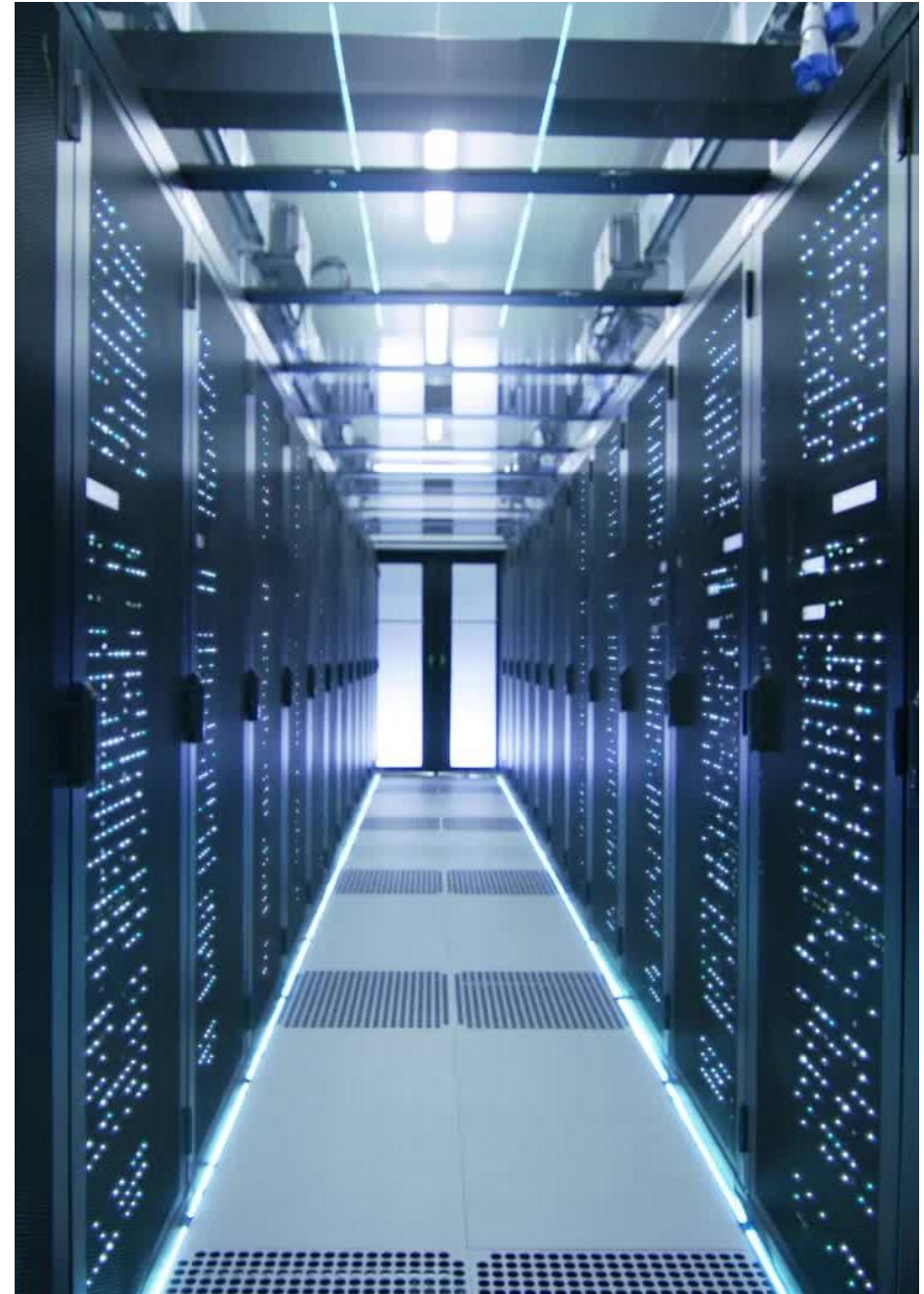
Lack of Change Management and Documentation

Risks of Unstructured Updates

Lack of change management causes sporadic updates increasing vulnerability to zero-day exploits and operational issues.

Security Risks from Direct Connections

Direct update connections to external servers raise risks like man-in-the-middle attacks compromising system integrity.



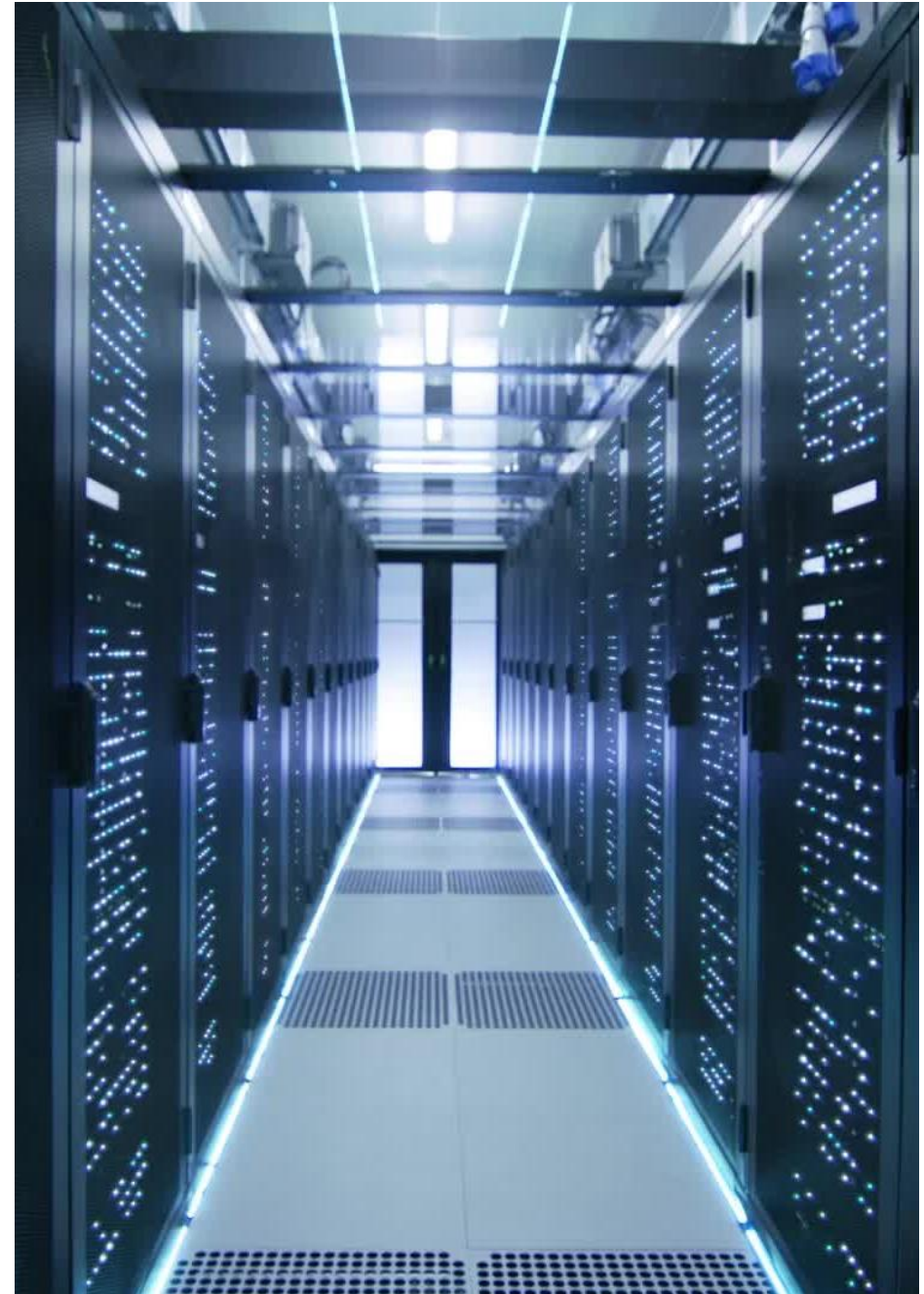
Lack of Change Management and Documentation

Benefits of WSUS Implementation

WSUS provides centralized, automated update control ensuring digital signature verification and reducing tampering risks.

Improved Compliance and Stability

Automating updates with WSUS enhances compliance, system compatibility, and strengthens overall security and stability.



Conclusion



Summary and Recommendations

Current Vulnerabilities

SecureWave faces risks from unstructured patching, direct updates, and outdated firmware, compromising security and continuity.

Patch Management Improvements

Implementing a structured patch management cycle and documented change control is essential to reduce exposure to threats.



Summary and Recommendations (Cont.)

Proactive Security Measures

Continuous monitoring, vulnerability scanning, and centralized updates via WSUS help minimize attack surfaces effectively.

Benefits of Recommendations

Adopting these strategies ensures infrastructure protection, regulatory compliance, and sustained trust with stakeholders.

End.

Thank You